1. What is abstract class explain with an example.

→ An abstract class in Java is a class that cannot be instantiated. It meant to be inherited by other classes.

Example:

```
abstract class Animal {
    abstract void makeSound();
    void sleep() {
System.out.println("Sleeping....");
    }
}

class Dog extends Animal {
    void makeSound() {
System.out.println("Dog barks");
    }
}

public class Main {
public static void main(String[]args) {
Animal obj = new Dog();
 obj.makeSound();
 obj.sleep();
    }
}
```

Output:

```
Dog barks
Sleeping....
```

2. Illustrate method overloading with an example

Method overloading is a feature in java that allows a class to have more than one method with the same name but different parameters.

Example :

```
class Demo {
    void show(int a) {
        System.out.println("int :" + a);
    }

    void show(String b) {
        System.out.println("String :" + b);
    }
}

public class Main {
    public static void main(String[] args) {
        Demo d = new Demo();
        d.show(10);
        d.show("Hello");
    }
}
```

Output :

```
int : 10
String : Hello.
```

3. Demonstrate the working of final class and final method in java

→ Final class

* ✱ class declared with final cannot be extended (no subclass can inherit it).

→ Final method

* ✱ method declared with final cannot be overridden in a subclass

Working example:

```
final class Calculator {
    final int square(int n) {
    return n * n;
    }
}
// It gives error if we try to extend Calculator.

class Printer {
    final void printMessage() {
System.out.println("Printing from printer");
    }
}

class ColorPrinter extends Printer {
    // Overriding this causes error.
}

public class Main {
public static void main(String[]args) {
    Calculator calc = new Calculator();
System.out.println("Square of 5: "+ calc.square(5));
    ColorPrinter cp = new
ColorPrinter();
        cp.printMessage();
}
```

```
}
```

Output :

```
Square of 5 : 25
Printing from printer.
```

4. What is exception give an example.

=> An exception is an unexpected event or error that occurs during the execution of a program, which disrupts the normal flow of instructions.

Example :

```
public class Main {
public static void main (String[]args) {
    try {
        int result = 10/0;
System.out.println("Result: " + result);
}
    catch (ArithmeticException e)
{
System.out.println("Exception caught:" +e);
}
}
}
```

Output :

```
Exception caught:
java.lang.ArithmeticException: / by zero.
```

5. Working of array index out of Bound & Exception.

=>
```
import java.util.Scanner;
public class ShortExample {
 public static void main (String[]args)
   Scanner sc = new Scanner (System.in);
  try {
      int a = sc.nextInt ();
      int b = sc.nextInt ();
       if (b==0)
      throw new
ArithmeticException ("b cannot be zero");
 System.out.println ("a/b = " + (a/b));
    int[] arr = {1,2,3};
System.out.println (arr[5]);
}
   catch (ArithmeticException
            ArrayIndexOutOfBoundsException e) {
System.out.println (" Exception: " + e);
}
    sc.close ();
  }
}
```

Sample Output:

```
10
2.
a/b = 5
Exception:
java.lang.ArrayIndexOutOfBoundsException:
Index 5 out of bounds for length 3
```

6. Explain the procedure of creating our own exception.

1. Extend the Exception or RuntimeException class.
   - Use exception if you want a checked exception
   - Use RuntimeException for an unchecked exception.

2. Create a constructor to pass a custom error message to the base class.
3. Throw the custom exception using the throw keyword.
4. Catch it using try-catch.

Example:

```
class MyException extends Exception {
public MyException (String message) {
  super(message);
  }
}
public class CustomExceptionDemo {
  public static void main (String[]args) {
  try {
    int age = 15;
    if (age < 18) {
    throw new
MyException ("Age must be 18 or above");
    }
  System.out.println ("Eligible to vote");
  } catch (MyException e) {
System.out.println ("Caught Custom Exception:" + e.get
                                    Message());
  }
  }
}
```

Output:

Caught Custom Exception: Age must be 18 or above